

Longitudinal Evaluation of a Multi-Site Residency Training Model

Association With Physician Practice Behavior and Early Clinical Outcomes

What happens after competency-based education?

Does it actually change how physicians practice—and ultimately affect patient care?

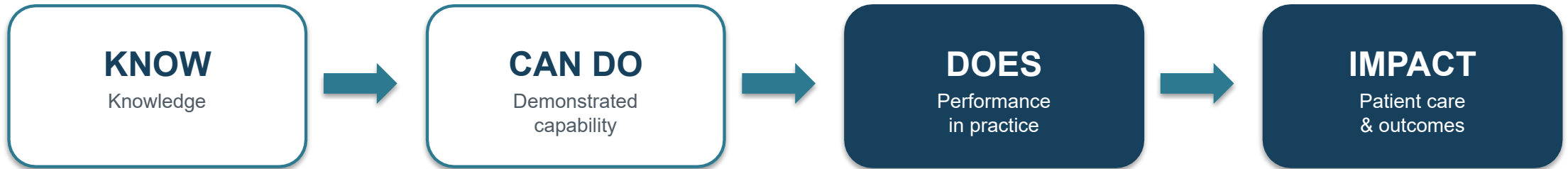
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Disclosure

Nothing to Disclose

Competence is more than knowledge



**Competency-based education is designed to change practice
But educational evaluation often stops before practice is measured.**

COMMONLY MEASURED
Knowledge • Confidence • Satisfaction

LESS COMMONLY MEASURED
Behavior in clinical practice

HARDEST TO DEMONSTRATE
Downstream patient & system effects

What happens when we follow competency development farther downstream?

The Lifestyle Medicine Residency Curriculum

WHAT IS THE MODEL?

40
didactic hours

+

60
applied hours

100-hour curriculum

PRACTICUM / CLINICAL APPLICATION

400

LM patient encounters

10 hrs

Intensive Therapeutic Lifestyle Change (ITLC)

10 hrs

Group facilitation

Embedded longitudinally within primary residency training

WHAT COMPETENCIES ARE BEING DEVELOPED?

- 1 Evidence-based lifestyle interventions
- 2 **Behavior-change counseling / motivational interviewing**
- 3 **Lifestyle prescriptions and SMART goals**
- 4 Patient engagement and shared decision-making
- 5 **Clinical workflow integration**
- 6 Interprofessional care

Repeated clinical application—not education alone.

From education to clinical impact

1	REACTION	Did residents find the training relevant and useful?
2	LEARNING	Did knowledge and confidence change?
3	BEHAVIOR	Did physicians actually practice differently?
4	RESULTS	What happened in patient care and practice?

MIXED METHODS

- Pre/post surveys
- Longitudinal follow-up
- Semistructured interviews
- Quantitative + thematic analysis

For this audience, I'm going to concentrate on Levels 3 and 4.

Competencies did not remain in the classroom

All 23 measured clinical behaviors improved after training • $p < .001$ • moderate-to-large effects

DISCUSS

Nutrition counseling

2.44 → 3.09

$r = .524$

Residents more frequently discussed a whole-food, predominantly plant-based diet.

PRESCRIBE

Physical activity

2.23 → 2.90

$r = .479$

Residents more frequently wrote lifestyle prescriptions for physical activity.

EQUIP

Sleep resources

2.00 → 2.89

$r = .570$

Residents more frequently provided written resources supporting restorative sleep.

“I ask about diet, sleep, and stress at every visit . . . it’s just part of my review of systems now.”

— Participant 8

What physicians observed in their patients

89.9%

**Body-weight
improvement**

107 of 119

84.9%

**Blood-pressure
improvement**

101 of 119

82.4%

**Blood-glucose
improvement**

98 of 119

73.9%

**Blood-lipid
improvement**

88 of 119

48.8%

Diabetes remission

61.2%

Improved quality of life

50.4%

Weight maintenance ≥ 1 year

Resident-reported observations; patient outcomes were not independently verified and do not establish causality.

Competency development influenced how physicians saw their role

PRACTICE INTEGRATION • PROFESSIONAL IDENTITY • SYSTEMS INFLUENCE • AGENCY

IDENTITY

“This is not just a skillset . . . it’s a philosophy I carry into every patient visit.”

— P8

AGENCY

“I want to teach LM wherever I go . . . it’s part of who I am now.”

— P9

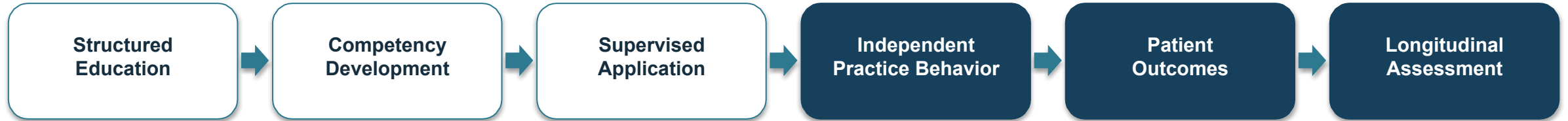
SYSTEMS

*“Our clinic flow changed to allow longer visits when LM is discussed.
That never would have happened without our involvement.”*

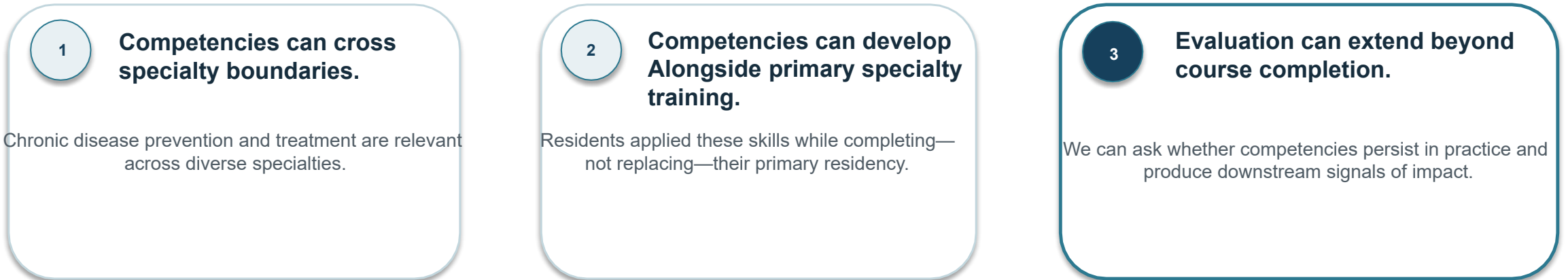
— P12

The change was not only what physicians did—it was how they understood their responsibility in chronic disease care.

What does competency-based education look like when we follow it downstream



THREE OBSERVATIONS FROM THIS STUDY



The question shifts from whether training occurred to whether competence persists in practice.

Competency development can be followed into clinical practice

